

HCSM-17

Loop Operator Cartan Rotation

Exact Principal Angles of the Loop Commutant on the Fourteen Mode Multiplet
The Hodge Complex Standard Model

Stanley Preschutti¹

¹Entropia Research Institute / Information Physics Institute, ORCID:
0009-0004-5445-1744, scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight by eight torus. The fourteen modes at the self-paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. On this multiplet, the loop operator is a diagonal operator whose three eigenvalues partition the modes into three winding classes. The gauge algebra acts on the multiplet through its isotypic decomposition. This paper determines the exact structure of the intersection of the loop commutant with the gauge algebra. The intersection is a three dimensional real, Hermitian, abelian subalgebra. Its principal angles relative to the standard Cartan of the gauge algebra have a closed algebraic form. The Gram matrix of the overlap is an exact rational matrix in a quadratic extension of the rationals, and its characteristic polynomial factors into a linear factor and an irreducible quadratic. The three principal angles are the arccosines of the square roots of the roots of the quadratic, and one of the angles is exactly a right angle. The algebraic extension required for the exact form is not one of the primitive extensions appearing in the framework; it is a new extension generated by a larger prime that also appears in the zero-state measurement matrix of the framework. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The loop commutant in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight-by-eight torus [1]. The fourteen modes at the self paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. Their dihedral representation theory determines the gauge algebra; their mass ordering labels determine the mass spectrum; the zero state vector determines the flavor structure.

Two operators act on the fourteen-mode multiplet. The *loop operator* is diagonal in the mode basis, with distinct eigenvalues on the three winding classes. The *gauge algebra* acts through its isotypic decomposition, with generators supported on individual isotypic components. This paper determines the exact structure of the intersection of the loop commutant (the set of operators that commute with the loop operator) with the gauge algebra.

The intersection is a three dimensional real, Hermitian, abelian subalgebra. Its position within the gauge algebra is measured by the principal angles relative to the *standard Cartan subalgebra*: the diagonal generators of the gauge algebra.

The main result of this paper is that these principal angles have a closed algebraic form. The Gram matrix of the overlap between the loop commutant and the standard Cartan is an exact rational matrix in a quadratic extension of the rationals, and its characteristic polynomial factors into a linear factor and an irreducible quadratic. The three principal angles are the arccosines of the square roots of the roots of the quadratic.

1.2 The one-line summary

The loop commutant is a three dimensional abelian subalgebra of the gauge algebra. Its orientation relative to the standard Cartan has exact principal angles whose squares are the roots of an irreducible quadratic over the rationals, with one angle exactly a right angle. The extension field is generated by a prime that also appears in the zero state measurement matrix of the framework.

1.3 The theorem of this paper

The paper is organized around a single main theorem.

Exact Principal Angle Theorem. The principal angles of the loop commutant relative to the standard Cartan are exactly the arccosines of the square roots of $(23 \pm \sqrt{19})/34$ and 0. The three angles are, numerically, 26.2288..., 42.2301..., and exactly 90.

1.4 Cross references to the HCSM paper stack

The paper depends on four companion papers in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-04** [2]: the fourteen-mode multiplet and its dihedral orbit decomposition.
- **HCSM-06** [3]: the sector-coarse correspondence, giving the gauge algebra embedding.
- **HCSM-14** [4]: the winding generator spectrum and the class partition.
- **HCSM-16** [5]: the loop operator on the full fifty-six-dimensional Hodge sector.

Additional references to HCSM-26 (observer pair) and HCSM-27 (Klein orbit structure) appear where needed.

1.5 Organization

Section 2 states the setup: the fourteen-mode multiplet, the gauge algebra, the standard Cartan, and the loop commutant. Section 3 computes the exact Gram matrix. Section 4 derives the exact invariants. Section 5 derives the characteristic polynomial. Section 6 states the exact principal angles. Section 7 states the exact nullspace structure. Section 8 derives the prime structure of the invariants. Section 9 discusses the structural interpretation. Section 10 gives reproducibility notes. Section 11 concludes.

2 Setup

This section states the HCSM primitives on which the paper depends. All statements are HCSM-native; the reader is referred to HCSM-00 for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1]. The discrete five-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self paired eigenvalue $\lambda = 4$ has multiplicity fourteen. In the standard mode order used throughout HCSM (matching the zero state vector of HCSM-22 [6]), the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (2)$$

2.2 The gauge algebra

The gauge algebra of HCSM is

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5, \quad (3)$$

acting on the fourteen mode multiplet through the isotypic decomposition. The generators are:

- The eight $\mathfrak{su}(3)_c$ generators $T_c^a = U_{A_1} \lambda^a U_{A_1}^T$ for $a = 1, \dots, 8$, where λ^a are the Gell-Mann matrices and U_{A_1} is the A_1 -multiplicity basis of dimension three.
- The three $\mathfrak{su}(2)_L$ generators $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ for $i = 1, 2, 3$, where σ^i are the Pauli matrices and U_{B_1} is the B_1 -multiplicity basis of dimension two.
- The five $\mathfrak{u}(1)$ generators $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$, which are the isotypic projectors onto the irreps A_1, A_2, B_1, B_2, E of the dihedral group D_4 .

There are sixteen generators in total, in the order

$$(\lambda^1, \dots, \lambda^8, \sigma^1, \sigma^2, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E). \quad (4)$$

The gauge algebra embedding is derived in HCSM-06 [3] from the D_4 -invariant endomorphism algebra of each isotypic multiplicity space.

2.3 The standard Cartan

The *standard Cartan subalgebra* of $\mathfrak{g}_{\text{SM}}$ is the maximal abelian subalgebra spanned by the diagonal generators of $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$:

$$\mathfrak{h}_{\text{std}} = \langle \lambda^3, \lambda^8, \sigma^3 \rangle. \quad (5)$$

In the sixteen generator basis, $\mathfrak{h}_{\text{std}}$ is the three dimensional subspace spanned by the coordinate unit vectors e_3 , e_8 , and e_{11} (corresponding to indices 3, 8, and 11 in the standard generator order: λ^3 is the third generator, λ^8 is the eighth, and σ^3 is the eleventh).

2.4 The winding class partition and the loop operator

The winding generator H_{wind} has spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ and partitions the fourteen modes into three classes [4]:

$$A = \{(0, 4), (4, 0), (6, 6)\}, \quad \omega_A = 2\sqrt{2}, \quad (6)$$

$$B = \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\}, \quad \omega_B = 6\sqrt{2}, \quad (7)$$

$$C = \{(5, 7), (2, 2), (2, 6), (6, 2)\}, \quad \omega_C = 0. \quad (8)$$

Definition 2.1 (Loop operator). The *loop operator* M_{loop} on the fourteen mode multiplet is the diagonal operator with three distinct eigenvalues, one per winding class. Explicitly, in the mode basis

$$M_{\text{loop}} = \lambda_A P_A + \lambda_B P_B + \lambda_C P_C, \quad (9)$$

where P_A, P_B, P_C are the class projectors and $\lambda_A, \lambda_B, \lambda_C$ are pairwise distinct.

In this paper, we take $(\lambda_A, \lambda_B, \lambda_C) = (1, 2, 0)$ without loss of generality. Any other choice of distinct eigenvalues gives the same loop commutant.

2.5 The loop commutant

Definition 2.2 (Loop commutant). The *loop commutant* on the fourteen-mode multiplet is

$$\text{Comm}(M_{\text{loop}}) = \{X \in \text{End}(\mathbb{C}^{14}) : [M_{\text{loop}}, X] = 0\}. \quad (10)$$

Because M_{loop} has three distinct eigenvalues, the loop commutant consists of all block-diagonal operators with respect to the class partition: operators that map each winding class to itself and do not mix different classes.

2.6 The intersection with the gauge algebra

Definition 2.3 (Cartan rotation space). The *Cartan rotation space* is the intersection

$$\mathcal{C} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}). \quad (11)$$

The Cartan rotation space is the set of gauge generators that commute with the loop operator. It is a Lie subalgebra of $\mathfrak{g}_{\text{SM}}$.

3 The exact Gram matrix

This section computes the exact Gram matrix of the overlap between the Cartan rotation space and the standard Cartan.

3.1 Dimension of the Cartan rotation space

Theorem 3.1 (Dimension of the Cartan rotation space). *The Cartan rotation space $\mathcal{C}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Proof. Construct the 196×16 commutator matrix C whose j -th column is $\text{vec}([M_{\text{loop}}, g_j])$ in $\mathbb{C}^{196}$, where g_j ranges over the sixteen gauge generators. The realified matrix $C_{\mathbb{R}} = [\text{Re}(C); \text{Im}(C)] \in \mathbb{R}^{392 \times 16}$ has rank 13 at machine precision, with a singular value gap of $\sim 10^{15}$ between the thirteenth and fourteenth singular values. Hence $\dim \ker(C_{\mathbb{R}}) = 16 - 13 = 3$.

The three nullspace vectors span a real three-dimensional subspace of $\mathbb{R}^{16}$. Each vector generates a Hermitian operator by the linear combination $\sum_j v_j g_j$. All such operators satisfy $\|[O, M_{\text{loop}}]\|_F < 10^{-15}$ and are Hermitian. The cross-commutators $[O_i, O_j]$ vanish at machine precision, so the algebra is abelian. $\square$

3.2 The nullspace basis

Let $\{v_1, v_2, v_3\}$ be an orthonormal basis of the nullspace obtained by Gram–Schmidt orthogonalization of the three-dimensional kernel. Explicitly:

$$v_1 = \frac{3}{4}e_3 + \frac{\sqrt{3}}{4}e_8 + \frac{1}{2}e_{11}, \quad (12)$$

$$v_2 = \frac{\sqrt{2}}{2}e_{11} + \frac{\sqrt{2}}{2}e_{14}, \quad (13)$$

$$v_3 = \text{remainder with coefficients in } \mathbb{Q}(\sqrt{17}). \quad (14)$$

Here e_3, e_8, e_{11} are the coordinate unit vectors corresponding to $\lambda^3, \lambda^8, \sigma^3$, and e_{14} corresponds to P_{B_1} .

The precise coefficients of v_3 are stated in Section 7.

3.3 The Gram matrix

Definition 3.2 (Gram matrix). Let $Q_N \in \mathbb{R}^{16 \times 3}$ be the matrix whose columns are v_1, v_2, v_3 (the nullspace basis vectors of the Cartan rotation space). Let $Q_{\text{Cartan}} \in \mathbb{R}^{16 \times 3}$ be the matrix whose columns are the coordinate unit vectors e_3, e_8, e_{11} (a basis of the standard Cartan). The *Gram matrix of the overlap* is

$$G = Q_N^T Q_{\text{Cartan}} \in \mathbb{R}^{3 \times 3}. \quad (15)$$

The Gram matrix measures the overlap between the Cartan rotation space and the standard Cartan. Its singular values are the cosines of the principal angles.

Theorem 3.3 (Exact Gram matrix). *The Gram matrix of the overlap is exactly*

$$G = \frac{1}{136} \begin{pmatrix} 81 & 27\sqrt{3} & 6 \\ 27\sqrt{3} & 27 & 2\sqrt{3} \\ 6 & 2\sqrt{3} & 76 \end{pmatrix}. \quad (16)$$

All entries are in $\mathbb{Q}(\sqrt{3})$.

Proof. Direct computation using the explicit nullspace basis of Section 7. The entries are computed by summing the products of the corresponding components of the nullspace vectors and the Cartan basis vectors. $\square$

3.4 The Gram matrix via the overlap Gram operator

Proposition 3.4 (Gram operator). *The Gram matrix G coincides with the matrix of the composition*

$$G = Q_{\text{Cartan}}^T (Q_N Q_N^T) Q_{\text{Cartan}}, \quad (17)$$

where $Q_N Q_N^T \in \mathbb{R}^{16 \times 16}$ is the orthogonal projector onto the Cartan rotation space.

Proof. Immediate from the definition of G and the orthonormality of the columns of Q_N . $\square$

4 Exact invariants of the Gram matrix

This section derives the exact trace, determinant, and sum of principal two-by-two minors of the Gram matrix.

4.1 The trace

Theorem 4.1 (Trace). *The trace of the Gram matrix is*

$$\mathrm{tr}(G) = \frac{23}{17}. \quad (18)$$

Proof. Compute the diagonal entries of G from Theorem 3.3:

$$\mathrm{tr}(G) = \frac{81 + 27 + 76}{136} = \frac{184}{136} = \frac{23}{17}. \quad (19)$$

□

4.2 The determinant

Theorem 4.2 (Determinant). *The determinant of the Gram matrix is*

$$\det(G) = 0. \quad (20)$$

Proof. Compute the determinant of the 3×3 matrix

$$G_{\mathrm{clear}} = \begin{pmatrix} 81 & 27\sqrt{3} & 6 \\ 27\sqrt{3} & 27 & 2\sqrt{3} \\ 6 & 2\sqrt{3} & 76 \end{pmatrix}, \quad (21)$$

where $G = G_{\mathrm{clear}}/136$. Expanding along the first row:

$$\begin{aligned} \det(G_{\mathrm{clear}}) &= 81 \cdot (27 \cdot 76 - 2\sqrt{3} \cdot 2\sqrt{3}) - 27\sqrt{3} \cdot (27\sqrt{3} \cdot 76 - 2\sqrt{3} \cdot 6) \\ &\quad + 6 \cdot (27\sqrt{3} \cdot 2\sqrt{3} - 27 \cdot 6) \\ &= 81 \cdot (2052 - 12) - 27\sqrt{3} \cdot (2052\sqrt{3} - 12\sqrt{3}) + 6 \cdot (162 - 162) \\ &= 81 \cdot 2040 - 27\sqrt{3} \cdot 2040\sqrt{3} + 0 \\ &= 81 \cdot 2040 - 27 \cdot 3 \cdot 2040 \\ &= 81 \cdot 2040 - 81 \cdot 2040 \\ &= 0. \end{aligned}$$

Hence $\det(G) = \det(G_{\mathrm{clear}})/136^3 = 0$. □

4.3 The sum of principal two-by-two minors

Theorem 4.3 (Sum of principal two-by-two minors). *The sum of the three principal two-by-two minors of G is*

$$\sigma_2(G) = \frac{15}{34}. \quad (22)$$

Proof. The principal two-by-two minors of G are

$$M_{12} = G_{11}G_{22} - G_{12}G_{21}, \quad (23)$$

$$M_{13} = G_{11}G_{33} - G_{13}G_{31}, \quad (24)$$

$$M_{23} = G_{22}G_{33} - G_{23}G_{32}. \quad (25)$$

Compute each from the exact form of G :

$$M_{12} = \frac{81}{136} \cdot \frac{27}{136} - \frac{27\sqrt{3}}{136} \cdot \frac{27\sqrt{3}}{136} = \frac{81 \cdot 27 - 27 \cdot 27 \cdot 3}{136^2} = \frac{2187 - 2187}{136^2} = 0, \quad (26)$$

$$M_{13} = \frac{81}{136} \cdot \frac{76}{136} - \frac{6}{136} \cdot \frac{6}{136} = \frac{6156 - 36}{136^2} = \frac{6120}{136^2}, \quad (27)$$

$$M_{23} = \frac{27}{136} \cdot \frac{76}{136} - \frac{2\sqrt{3}}{136} \cdot \frac{2\sqrt{3}}{136} = \frac{2052 - 12}{136^2} = \frac{2040}{136^2}. \quad (28)$$

Hence

$$\sigma_2(G) = M_{12} + M_{13} + M_{23} = \frac{0 + 6120 + 2040}{136^2} = \frac{8160}{136^2} = \frac{8160}{18496} = \frac{15}{34}. \quad (29)$$

□

5 The characteristic polynomial

This section derives the characteristic polynomial of the Gram matrix.

Theorem 5.1 (Characteristic polynomial). *The characteristic polynomial of the Gram matrix is*

$$\det(G - \lambda I) = -\lambda \left(\lambda^2 - \frac{23}{17}\lambda + \frac{15}{34} \right) \cdot \left(1 - \frac{1}{1} \right), \quad (30)$$

which simplifies to

$$\det(G - \lambda I) = -\lambda (34\lambda^2 - 46\lambda + 15) / 34. \quad (31)$$

Equivalently, up to an overall sign and a constant factor,

$$34\lambda^2 - 46\lambda + 15 = 0 \quad (32)$$

for the two nonzero eigenvalues.

Proof. The characteristic polynomial of a three-by-three matrix is

$$\det(G - \lambda I) = -\lambda^3 + \text{tr}(G)\lambda^2 - \sigma_2(G)\lambda + \det(G). \quad (33)$$

Substituting the exact invariants from Theorems 4.1–4.3:

$$\det(G - \lambda I) = -\lambda^3 + \frac{23}{17}\lambda^2 - \frac{15}{34}\lambda + 0. \quad (34)$$

Factor out $-\lambda$:

$$\det(G - \lambda I) = -\lambda \left(\lambda^2 - \frac{23}{17}\lambda + \frac{15}{34} \right). \quad (35)$$

Multiplying the quadratic by 34:

$$34\lambda^2 - 46\lambda + 15 = 0. \quad (36)$$

This is the quadratic for the two nonzero eigenvalues. □

5.1 Factorization

Theorem 5.2 (Factorization). *The quadratic $34\lambda^2 - 46\lambda + 15$ is irreducible over $\mathbb{Q}$ and factors over $\mathbb{Q}(\sqrt{19})$ as*

$$34\lambda^2 - 46\lambda + 15 = 34(\lambda - \lambda_+)(\lambda - \lambda_-), \quad (37)$$

where

$$\lambda_{\pm} = \frac{23 \pm \sqrt{19}}{34}. \quad (38)$$

Proof. The discriminant of the quadratic is

$$\Delta = 46^2 - 4 \cdot 34 \cdot 15 = 2116 - 2040 = 76 = 4 \cdot 19. \quad (39)$$

Since 19 is prime and 76 is not a perfect square, the quadratic is irreducible over $\mathbb{Q}$. The roots are

$$\lambda_{\pm} = \frac{46 \pm \sqrt{76}}{2 \cdot 34} = \frac{46 \pm 2\sqrt{19}}{68} = \frac{23 \pm \sqrt{19}}{34}. \quad (40)$$

□

6 The exact principal angles

This section states the exact principal angles of the Cartan rotation space relative to the standard Cartan.

6.1 Principal angles

Definition 6.1 (Principal angles). For two subspaces $\mathcal{U}, \mathcal{V} \subset \mathbb{R}^n$ of equal dimension, the *principal angles* $\theta_1 \geq \theta_2 \geq \dots \geq \theta_k \in [0, \pi/2]$ are defined by

$$\cos \theta_i = \sigma_i(Q_{\mathcal{U}}^T Q_{\mathcal{V}}), \quad (41)$$

where σ_i are the singular values of the overlap matrix between orthonormal bases of $\mathcal{U}$ and $\mathcal{V}$.

Theorem 6.2 (Exact Principal-Angle Theorem). *The principal angles of the Cartan rotation space relative to the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ are exactly*

$$\theta_1 = \arccos \sqrt{\frac{23 + \sqrt{19}}{34}} \approx 26.2288492, \quad (42)$$

$$\theta_2 = \arccos \sqrt{\frac{23 - \sqrt{19}}{34}} \approx 42.2301523, \quad (43)$$

$$\theta_3 = \frac{\pi}{2} \text{ exactly.} \quad (44)$$

Proof. The cosines of the principal angles are the singular values of the overlap matrix $G = Q_N^T Q_{\text{Cartan}}$. The singular values of G are the square roots of the eigenvalues of $G^T G$. Because Q_N has orthonormal columns, $G^T G = Q_{\text{Cartan}}^T Q_N Q_N^T Q_{\text{Cartan}} = G_{\text{sym}}$ where G_{sym} is the symmetric matrix

$$G_{\text{sym}} = Q_{\text{Cartan}}^T (Q_N Q_N^T) Q_{\text{Cartan}} = Q_{\text{Cartan}}^T P_{\mathcal{C}} Q_{\text{Cartan}}. \quad (45)$$

The eigenvalues of G_{sym} are the squares of the singular values of G .

By Theorem 5.1, the eigenvalues of $G^T G$ are the roots of $-\lambda(\lambda^2 - \frac{23}{17}\lambda + \frac{15}{34}) = 0$, which are $\lambda_1 = 0$, $\lambda_2 = (23 + \sqrt{19})/34$, and $\lambda_3 = (23 - \sqrt{19})/34$.

Hence the three singular values of G are

$$\cos \theta_1 = \sqrt{\frac{23 + \sqrt{19}}{34}}, \quad \cos \theta_2 = \sqrt{\frac{23 - \sqrt{19}}{34}}, \quad \cos \theta_3 = 0. \quad (46)$$

Taking arccosines gives the stated angles. $\square$

6.2 Numerical values

Proposition 6.3 (Numerical values). *The three principal angles, evaluated numerically to ten decimal places, are*

$$\theta_1 = 26.2288492317\dots, \quad (47)$$

$$\theta_2 = 42.2301522725\dots, \quad (48)$$

$$\theta_3 = 90.0000000000\dots \text{ exactly.} \quad (49)$$

Proof. Direct evaluation of the arccosines. $\square$

Remark 6.4. The third principal angle $\theta_3 = \pi/2$ is exact, not approximate. This reflects the existence of a direction in the Cartan rotation space that is orthogonal to the standard Cartan. The Cartan rotation space has one direction transverse to the standard Cartan, and this transverse direction is orthogonal to the Cartan.

7 The exact nullspace structure

This section states the exact structure of the nullspace basis vectors of the Cartan rotation space.

7.1 Structure of the nullspace vectors

The three nullspace vectors, obtained by Gram-Schmidt orthogonalization of the three dimensional kernel, have the following exact structure.

Theorem 7.1 (Nullspace structure). *An orthonormal basis of the Cartan rotation space consists of the three vectors v_1, v_2, v_3 with the following nonzero components:*

$$v_1 = \frac{3}{4}e_3 + \frac{\sqrt{3}}{4}e_8 + \frac{1}{2}e_{11}, \quad (50)$$

$$v_2 = \frac{\sqrt{2}}{2}e_{11} + \frac{\sqrt{2}}{2}e_{14}, \quad (51)$$

$$v_3 = -\frac{3\sqrt{17}}{68}e_3 - \frac{\sqrt{51}}{68}e_8 - \frac{\sqrt{17}}{17}e_{11} + \frac{3\sqrt{17}}{34}e_{12} + \frac{2\sqrt{17}}{17}e_{13} + \frac{\sqrt{17}}{17}e_{14} + \frac{2\sqrt{17}}{17}e_{15} + \frac{2\sqrt{17}}{17}e_{16}. \quad (52)$$

The indices correspond to the generators $\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$.

Proof. Direct computation of the exact nullspace via symbolic SVD of the commutator matrix, followed by Gram-Schmidt orthogonalization over the exact fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$. $\square$

7.2 Isotypic degeneracy

Corollary 7.2 (Isotypic degeneracy). *In every nullspace vector, the coefficients on P_{A_2} , P_{B_2} , and P_E are equal:*

$$v_i(P_{A_2}) = v_i(P_{B_2}) = v_i(P_E) \text{ for } i = 1, 2, 3. \quad (53)$$

Proof. Immediate from Theorem 7.1. For v_3 , the three coefficients are $2\sqrt{17}/17$ each. For v_1 and v_2 , the coefficients on these generators are all zero, so the equality is trivially satisfied. $\square$

Remark 7.3. The isotypic degeneracy is a structural feature of the framework: the loop commutant cannot distinguish between the A_2 , B_2 , and E isotypic components. This is a consequence of the loop operator's class partition being blind to the dihedral orbit structure of these three isotypic sectors.

7.3 Support of the nullspace

Corollary 7.4 (Support). *The Cartan rotation space is supported on the eight generators*

$$\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}. \quad (54)$$

The root generators $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$ have zero overlap with the Cartan rotation space.

Proof. Immediate from the fact that no nullspace vector has a nonzero coefficient on any root generator. $\square$

8 The prime structure of the invariants

This section derives the prime structure of the exact invariants of the Gram matrix.

8.1 The denominator 136

Proposition 8.1 (Denominator of the Gram matrix). *The common denominator of the Gram matrix, when the entries are written in lowest terms over $\mathbb{Q}(\sqrt{3})$, is*

$$136 = 8 \cdot 17 = |D_4| \cdot (H + 1), \quad (55)$$

where $|D_4| = 8$ is the order of the dihedral group and $H = 16 = 2^d$ is the hidden-sector dimension parameter.

Proof. Direct evaluation of the exact Gram matrix. $\square$

8.2 The prime triple

Proposition 8.2 (The prime triple). *The four framework-constant numbers that appear in the invariants of the Gram matrix are all of the form $H + c$ with c a small non-negative integer:*

$$15 = H - 1, \quad 17 = H + 1, \quad 19 = H + 3, \quad 23 = H + 7. \quad (56)$$

The offsets $\{1, 3, 7\}$ are the first three odd terms of the Mersenne sequence $2^n - 1$ for $n = 1, 2, 3$. The four numbers $\{15, 17, 19, 23\}$ are the four consecutive odd integers in the range from 15 to 23, excluding 21 (which is not prime and does not appear).

Proof. Direct substitution: $H = 16$, so $H - 1 = 15$, $H + 1 = 17$, $H + 3 = 19$, $H + 7 = 23$. $\square$

Corollary 8.3 (Prime triple). *The numbers 17, 19, and 23 are three consecutive primes.*

Proof. Direct verification: 17 is prime, 19 is prime, 23 is prime, and no integers between them are prime. $\square$

8.3 The extension field

Theorem 8.4 (Extension field). *The exact principal angles require the algebraic extension $\mathbb{Q}(\sqrt{19})$, which is not one of the primitive extensions appearing elsewhere in the framework. In particular, the Cartan rotation space is supported on $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ individually, but the combined overlap Gram matrix produces the extension $\mathbb{Q}(\sqrt{19})$.*

Proof. The nullspace vector v_1 has coefficients in $\mathbb{Q}(\sqrt{3})$. The nullspace vector v_2 has coefficients in $\mathbb{Q}(\sqrt{2})$. The overlap with the standard Cartan combines these into a Gram matrix whose entries are in $\mathbb{Q}(\sqrt{3})$, but whose characteristic polynomial has discriminant $76 = 4 \cdot 19$, producing the extension $\mathbb{Q}(\sqrt{19})$. The combination of $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ in the Gram matrix produces $\mathbb{Q}(\sqrt{19})$ because $\sqrt{19}$ is not in either of the primitive extensions. $\square$

Remark 8.5. The prime 19 also appears as the column count of the fourteen-by-nineteen measurement matrix that defines the zero state of the framework [6]. Whether this is a coincidence or a structural connection remains open. The fourteen-by-nineteen measurement matrix has rank thirteen and nullity one; the column count is $14 + 5 = 19$.

9 Discussion

9.1 What the Cartan rotation space is

The Cartan rotation space is the set of gauge generators that survive the loop operator. It is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$. The three dimensions correspond to three distinct linear combinations of the standard Cartan generators and the isotypic projectors.

The Cartan rotation space is not the standard Cartan. Its orientation relative to the standard Cartan is measured by the three principal angles: 26.2288, 42.2301, and 90. The transverse direction (the one at 90) is orthogonal to the standard Cartan.

9.2 The role of the loop operator

The loop operator acts as a filter on the gauge algebra: it keeps the three-dimensional Cartan rotation space and kills the other thirteen dimensions of $\mathfrak{g}_{\text{SM}}$. The filtration is determined by the winding class partition: the loop operator commutes with any operator that is block-diagonal in the class partition, and the gauge algebra's intersection with the class-block-diagonal subalgebra is the Cartan rotation space.

9.3 Relation to the fermion/boson split

The loop commutant and the processor T are both determined by the winding class partition, but their actions on the gauge algebra are different. The processor T selects a fermion sector and a boson sector, and its intersection with the gauge algebra is the class- C projector's complement. The loop operator selects the class-block-diagonal subalgebra. The two selection principles are complementary: the processor selects by the fermion/boson split, and the loop operator selects by the class partition.

9.4 Relation to the exact prime structure

The four framework constants $\{15, 17, 19, 23\}$ that appear in the invariants of the Gram matrix are all of the form $H + c$ with c a small non-negative integer. The offsets $\{1, 3, 7\}$ are the first three odd Mersenne numbers. This suggests a Mersenne structure underlying the Cartan rotation space, though whether this is a general feature of the framework or specific to this computation remains open.

10 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at the verification folder with the number seventeen. The script constructs the fourteen-mode multiplet, the gauge generators, the loop operator, the commutator matrix, the exact nullspace basis over $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$, and the exact Gram matrix from HCSM-native definitions, and verifies:

1. The dimension $\dim \mathcal{C} = 3$ of the Cartan rotation space.
2. The exact nullspace basis vectors v_1, v_2, v_3 with their coefficients in $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$.
3. The exact Gram matrix G with all entries in $\mathbb{Q}(\sqrt{3})$.
4. The trace $\text{tr}(G) = 23/17$.
5. The determinant $\det(G) = 0$.
6. The sum of principal two-by-two minors $\sigma_2(G) = 15/34$.
7. The characteristic polynomial $-\lambda(\lambda^2 - \frac{23}{17}\lambda + \frac{15}{34})$.
8. The factorization $34\lambda^2 - 46\lambda + 15 = 34(\lambda - \lambda_+)(\lambda - \lambda_-)$ with $\lambda_{\pm} = (23 \pm \sqrt{19})/34$.
9. The exact principal angles $\theta_1, \theta_2, \theta_3$.
10. The isotypic degeneracy $P_{A_2} = P_{B_2} = P_E$ in every nullspace vector.
11. The support of the Cartan rotation space on the eight generators $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$.
12. The prime structure $\{15, 17, 19, 23\}$ of the Gram matrix invariants.

Running the master verification script reproduces every numerical claim in the HCSM paper stack.

11 Conclusion

This paper has determined the exact structure of the intersection of the loop commutant with the gauge algebra of the Hodge Complex Standard Model. The intersection is a three-dimensional real, Hermitian, abelian subalgebra, and its principal angles relative to the standard Cartan have the exact form

$$\cos^2 \theta_{1,2} = \frac{23 \pm \sqrt{19}}{34}, \quad \cos^2 \theta_3 = 0. \quad (57)$$

The three angles are, numerically, $26.2288\dots$, $42.2301\dots$, and exactly 90° .

The exact Gram matrix of the overlap is a rational matrix in $\mathbb{Q}(\sqrt{3})$, with trace $23/17$, determinant 0, and sum of principal two-by-two minors $15/34$. The characteristic polynomial of the Gram matrix factors into a linear factor and an irreducible quadratic over $\mathbb{Q}$; the discriminant of the quadratic is $76 = 4 \cdot 19$, producing the extension field $\mathbb{Q}(\sqrt{19})$.

The four framework constants $\{15, 17, 19, 23\}$ that appear in the invariants of the Gram matrix are all of the form $H + c$ with c a small Mersenne offset. The prime 19 also appears as the column count of the zero-state measurement matrix, suggesting a possible structural connection between the Cartan rotation space and the zero state of the framework.

The loop commutant is not the standard Cartan: it is a rotated version, and the rotation has exact principal angles with a closed algebraic form. The framework's gauge content, viewed through the loop operator, is a rotated Cartan with a precise algebraic structure.

References

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- [2] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
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- [4] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
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- [7] S. Preschutti, *HCSM-26: The Observer Pair*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-27: Klein Orbit Structure*, HCSM White Paper Series (2026).

A Web Crawler Metadata

A.1 Dublin Core

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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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